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Studierichting: Informatica
Oefeningenreeks 5A nr. 9.1

Bereken de volgende bepaalde integraal door substitutie:

$$\int_0^1 (1-x^2)\cdot x^{\frac{1}{3}}\,dx$$

$$u = x^2 \Rightarrow du = 2xdx \Rightarrow dx = \frac{du}{2x}$$

$$= \int_0^1 (1-u)\cdot u^{\frac{1}{6}} \cdot \frac{du}{2\sqrt{u}}$$

$$= \int_0^1 (1-u)\cdot u^{\frac{1}{6}} \cdot \frac{1}{2\cdot u^{\frac{1}{2}}}\,du$$

$$= \frac{1}{2}\cdot \int_0^1 (1-u)\cdot u^{\frac{-1}{3}}\,du$$

$$= \frac{1}{2}\cdot \left(\int_0^1 u^{\frac{-1}{3}}\,du - \int_0^1 u^{\frac{2}{3}}\,du\right)$$

$$\int_0^1 u^{\frac{-1}{3}}\,du = \left[\frac{u^{\frac{2}{3}}}{\frac{2}{3}}\right]_0^1 = \frac{3}{2}$$

$$\int_0^1 u^{\frac{2}{3}}\,du = \left[\frac{u^{\frac{5}{3}}}{\frac{5}{3}}\right]_0^1 = \frac{3}{5}$$

$$\frac{1}{2}\cdot\left(\frac{3}{2}-\frac{3}{5}\right) = \frac{1}{2}\cdot\left(\frac{15-6}{10}\right) = \frac{9}{20}$$

Antwoord: $\boxed{\frac{9}{20}}$